

External manufacturing package

“Field Ambience Source files”

Package	Field Ambience Source files.zip — standalone KiCad 9.0.4 project “FIELD AMBIENCE”, gerbers, BOM.xlsx, CPL.xlsx, handoff PDF, 2 screenshots
Board	198.84 × 106.92 mm · 4 layers · 1.6 mm · 195 footprints · 239 nets · 166 vias
Audit date	2026-07-30
Reference	AMBIENT repo state r19.42 · kicad/generate_kicad_project.py · firmware-c-next/src/ha1_h743/
Verdict	DO NOT FABRICATE — 7 blockers, 15 important findings, 7 regressions

Method

The netlist was reconstructed from the pad-to-net assignments inside `.kicad_pcb` (195 footprints, 239 nets, 1462 track segments, 166 vias) and symbol pinouts were extracted from `.kicad_sch` (302 symbols). Every pin assignment was then mirrored against two independent references that already exist in this project: the datasheet-verified symbol definitions in `kicad/generate_kicad_project.py` (TI SLUS810N, SLVSD38C, SLVSD76C; Diodes AP7361C and PAM8406) and the running firmware in `firmware-c-next/src/ha1_h743/`. On top of that: numeric geometry checks (zone-to-pad clearance measured across 583 pad instances, component spacing), gerber / drill / job-file inspection, and a designator cross-check between PCB, BOM and CPL.

This layout was NOT produced by `kicad/generate_kicad_project.py`.

It is a separate, hand-drawn project built on EasyEDA-imported libraries. That matters for reading the findings below: it also falls behind several decisions this project had already closed and verified (see §3, Regressions). `PCB_LAYOUT_STATUS.md` still claimed “no `.kicad_pcb` exists” (dated 2026-06-11) — that is now corrected in the repo.

1 · Blockers — do not fabricate

Seven findings that either destroy hardware on first power-up, or leave a populated part permanently non-functional. Two of them (B1, B2) sit in the same 5 V branch and should be fixed as one re-layout.

B1	L1 is a 1.8 μ H 0402 chip inductor in the boost power path
Finding	L1 = MLI1005F-1R8KT (LCSC C2885840), footprint L0402 — 0.54 × 0.54 mm pads — sitting on the switch node of U1 (TPS61089).
Why it matters	U1 boosts VBAT (3.0–4.2 V) to 5 V. At only 500 mA of 5 V load the peak inductor current already exceeds 2 A. A 0402 multilayer chip inductor is specified for a few hundred milliamps and saturates far below that; its DCR is in the hundreds of milliohms.
Failure mode	The converter never reaches regulation, sits in permanent current limit, and the inductor is thermally destroyed. The entire 5 V branch hangs off this — amplifier, headphone amp, and the 3.3 V LDO that feeds the MCU.
Fix	Use the part already in the repo BOM: SWPA6045S2R2NT , 2.2 μ H, 6045, ~4 A (C36500). While doing so, re-check the type-II compensation (R3/C6) — it was dimensioned for 2.2 μ H, not 1.8 μ H.
B2	The entire boost input current runs through slide switch SW2
Finding	VSYS → SW2.1 / SW2.2 → PWR_ON, and PWR_ON feeds L1.1 and U1.9 (VIN) as well as U1.7 (EN) and U5.3 (ON).
Why it matters	SW2 is an SSSS811101 (C109335), an ALPS signal-level slide switch rated by LCSC at 12 V / 50 mA. 1.5–2 A flow through it here.
Failure mode	The contact welds shut or burns open — the unit ends up either permanently on or permanently dead. On top of that, a mechanically bouncing contact now sits directly in a switching converter's input path, which will restart the converter on every actuation.
Fix	Adopt the topology from ADR-0016 : VSYS goes directly to U1.VIN and the input capacitor. SW2 switches only the PWR_ON signal (EN of U1 plus ON of U5).
B3	PWR_ON has no pull-down resistor
Finding	SW2 is SPDT and its second throw (pad 3) is unconnected. Net PWR_ON carries only C2, L1.1, U1.7 (EN), U1.9 (VIN) and U5.3 (ON) — no resistor to GND .
Why it matters	With the switch open, the boost EN and the load-switch ON inputs are left floating.
Failure mode	The device switches itself on from coupled noise or oscillates around the enable threshold; there is no defined microamp quiescent state when “off”.
Fix	R_PWR_PD (100 k) from PWR_ON to GND — same edit pass as B2.
B4	USB-C: D+/D– wired for one plug orientation only
Finding	USBC1.B6 = DR+, USBC1.B7 = DR–; pads A6 (DP1) and A7 (DN1) are unconnected — the PCB reports them as unconnected- (USBC1-DP1-PadA6) and -DN1-PadA7.
Why it matters	On a USB 2.0 receptacle A6 must be tied to B6 and A7 to B7, otherwise only one of the two plug orientations carries data.
Failure mode	USB — DFU flashing, charger detection, host communication — fails to enumerate in roughly half of all insertions. This is exactly the class of fault that costs days of bring-up time because it reads as a “flaky board”.
Fix	Connect A6 to B6 and A7 to B7 at the connector.

B5**LED17 is wired backwards**

Finding	The XL-1608U0C-06 symbol defines pin 1 = K, pin 2 = A . On the board: LED17.1 (K) → R45 → +5V and LED17.2 (A) → PCA9685 channel 9.
Why it matters	All 16 other LEDs are correctly A → R → +5 V with K → PCA sink — including LED14–16, which are the same package family (XL-1608UWC-04) with the identical pin convention. LED17 is the only one reversed.
Failure mode	LED17 never lights. There is no bodge without cutting a trace.
Fix	Flip LED17 in the schematic.

B6**PCA9685 drives LEDs against +5 V while firmware selects totem-pole**

Finding	U11 runs from 3.3 V (pin 28) and the LED anodes sit on the un-gated +5 V rail through 390 Ω. <code>mcp23017_h743.c:197</code> writes <code>PCA_MODE2 = PCA_MODE2_OUTDRV</code> .
Why it matters	The PCA9685 LEDn outputs are 5.5 V tolerant only in open-drain mode . In totem-pole mode the absolute maximum is $V_{DD} + 0.5\text{ V} = 3.8\text{ V}$, and the 5 V pull-up continuously back-feeds the high-side driver and its body diode into the 3.3 V rail.
Failure mode	Latch-up risk on the PCA9685, LEDs glowing when nominally off, and injection into the 3.3 V supply.
Fix	Either set <code>MODE2 = 0x00</code> (open-drain) in <code>pca_dev_init()</code> , which makes the existing wiring datasheet-compliant, or move the LED anode rail to 3.3 V and re-compute the series resistors. The firmware one-liner is the cheap path, but it has to land before bring-up.

B7**MIDI jack CN5 lands on the wrong contacts**

Finding	CN2 and CN5 are the same part (PJ-320D, C431535) with the same footprint. On CN2 , GND is on pad 3 , audio L/R on pads 1/2, and the switch contact (jack detect) on pad 4. On CN5 , GND sits on pad 1 , R50 (10 k to 3.3 V) on pad 2, R49 (220 Ω from USART2_TX) on pad 4 , and pad 3 is unconnected .
Why it matters	Two identical parts carry contradictory pin assignments — at least one of the two jacks is wired to the wrong contacts. Independently of that, the CN5 network is not a MIDI TRS type-A output: the “+” leg needs roughly 33 Ω to 3.3 V (CA-033), not 10 kΩ. A 10 kΩ source cannot drive the 5 mA current loop of a MIDI opto-coupler.
Failure mode	MIDI out dead; worst case on CN2, audio appears on the switch contact instead of tip/ring.
Fix	Pin down the PJ-320D pad-to-contact mapping against the datasheet (flagged UNVERIFIED — NEEDS HUMAN CHECK), then rewire both jacks consistently. MIDI: tip ← 33 Ω ← USART2_TX, ring ← 33 Ω ← 3.3 V, sleeve = GND (repo reference r18.67 / ADR-0004).

2 · Important — resolve before fabrication

#	Finding	Impact	Fix
I1	STM32 VBAT (pin 6) is unconnected	ST requires VBAT tied to VDD when no backup cell is fitted. The backup domain and RTC are unreliable and start-up problems are possible — note that X2 (32.768 kHz LSE) is populated.	VBAT → 3.3 V + 100 nF
I2	VREF+ (pin 20) only decoupled by C43/C44, not tied to VDDA	The ADC has no reference; the pin floats unless firmware enables the internal VREFBUF.	VREF+ to the VDDA net, as the repo generator does
I3	Encoders 3 and 4 sit on pins with no timer quadrature — SW14 → PC4/PC5, SW15 → PB0/PB1	On the H743, PC4/PC5 have no TIM alternate function at all, and PB0/PB1 only offer CH3/CH4/CHxN — no TIMx_CH1/CH2 . The firmware runs all four encoders in TIM encoder mode (TIM2/3/4/1). Two of them would have to move to EXTI/polling; lost steps on fast rotation plus extra ISR load next to the audio hot path.	Move to PA0/PA1, PC6/PC7, PD12/PD13, PA8/PA9 (or the TIM2/TIM3 remaps PA15/PB3, PB4/PB5)
I4	Pin-map collisions with hal_h743 (detail in §4)	The board is not drop-in for the current firmware.	Plan a firmware port, or align the layout
I5	8 of 15 status LEDs are on a different PCA9685 channel than their net name implies	leds.c writes “LED n → channel n”, so the wrong LED lights.	Remap in firmware or fix the schematic (§5)
I6	Layer stack deviates from SPEC §9 (Sig/GND/PWR/Sig)	In1.Cu is a 3.3 V pour that is additionally sliced by 233 signal segments ; In2.Cu has no pour at all , only 151 tracks. The only ground is the F.Cu/B.Cu pours, which routing cuts apart. For a 480 MHz H7 plus a 2 MHz boost converter plus analogue audio, this is the single largest EMC risk in the layout.	Continuous GND plane on an inner layer; move signals off the plane layer
I7	DRC is effectively disabled in the project file	clearance, annular_width, hole_clearance, padstack, *_courtyard, silk_* and starved_thermal are all set to ignore , plus min_clearance = 0, min_track_width = 0, min_connection = 0. A DRC run against this project file proves nothing.	Restore severities to error and re-run DRC
I8	NPTH holes exported as plated holes with zero annular ring	The NPTH drill file is empty ; every mounting and anchor hole (CN2/CN5 1.0 mm, SW1/SW3–6 2.0 and 3.5 mm, SW2 0.9 mm, USBC1 0.75 mm) is in the PTH file with pad diameter equal to drill diameter.	Convert those pads to np_thru_hole in the footprints and re-export; otherwise expect a JLC DFM query
I9	Boost input capacitor 6.2 mm away from U1 (C2, 22 µF; output bank 4.6–9 mm)	The hot loop is far too large for a converter switching up to 2.4 MHz — overshoot on the switch node and EMC problems.	Re-place the power stage (already required by B1/B2)
I10	HSE crystal X1 on the opposite board side	~10 mm from PH0/PH1 and placed in the middle of the front panel. Electrically: a long, via-stitched clock path. Mechanically: an 11.4 × 4.7 mm HC49-SMD package sitting between the keys on the user-facing surface.	X1 plus C54/C55 directly at PH0/PH1 on the MCU side
I11	No series damping resistors on the LCD SPI	H1 is 54.5 mm from the MCU plus a cable to the module — ringing and EMI at the refresh rates the LCD-motion work targets.	22–33 Ω in series on SCK/MOSI/CS/DC near the MCU
I12	H1 pin order is non-standard (1 = VCC, 2 = GND, 3 = DIN, 4 = CLK, 5 = CS, 6 = DC, 7 = RST, 8 = BL)	Common ST7789 modules are GND, VCC, SCL, SDA, RES, DC, CS, BLK — plugging a stock module straight in puts VCC on GND. The backlight circuit (low-side 2N7002 on BL) also assumes BL is the LED cathode , whereas many modules expect BL to be an active-high enable.	Exact display module: UNVERIFIED — NEEDS HUMAN CHECK , then fix pinout and backlight path

#	Finding	Impact	Fix
I13	Battery NTC R13 is on the PCB, not on the cell	R13 (NCP15XH103) sits in the power corner next to the charger and the boost converter, so the BQ24074 TS input measures board temperature. Under load this can trip a temperature fault and abort charging. The repo deliberately uses a fixed 10 k resistor instead of an NTC.	Fixed 10 k, or a pack NTC at the cell via CN1
I14	PCA9685 EXTCLK (pin 25) floats	CMOS input; NXP specifies it must not be left open.	Tie to VSS
I15	PSRAM exposed pad (U9 pad 9) unconnected	Thermal and EMC.	Tie to GND

3 · Regressions against the repo state (r19.37 ff.)

Decisions this project had already made, verified and documented, which the external layout does not carry.

#	Board	Repo state	Consequence
R1	U8 = PAM8403DR (C17337)	PAM8406DR (C86270), r19.37 / ADR-0025 — the PAM8403 is NRND	Beyond the lifecycle question: the PAM8403 has a fixed +24 dB gain. The r19.37 gain-staging fix (R1 20 k → 174 k, +23 dB → +4.3 dB) cannot be applied to this board at all — the 5 V BTL amp clips in the analogue domain far below DAC full scale. This is precisely the bug that was already found and fixed.
R2	C69/C70 = 1 µF input coupling caps	10 nF (speaker HPF ~91 Hz, r19.37)	The high-pass now sits near 8 Hz; the 8 Ω 40 mm driver receives the full bass energy.
R3	TPA6132A2 gain set to 0 dB (G0 = H, G1 = L)	−6 dB per ADR-0024	Re-assess the level plan — together with R1.
R4	U3 = AP7361C-33E-13, SOT-223 (C500795), wired as 1 = IN, 2 = GND, 3 = OUT, tab = GND	AP7361C-33Y5-13, SOT-89-5 (C460397); r18.6 explicitly rejected a SOT-223 variant because its pinout differs	AP7361C SOT-223 pin order: UNVERIFIED — NEEDS HUMAN CHECK. If wrong, the entire 3.3 V rail is dead. Must be checked against the Diodes datasheet before fabrication.
R5	U9 = APS6404L-3SQR-ZR, USON-8 (C3040877)	APS6404L-3SQN-SN, SOIC-8 (C3028887)	Pin mapping checked and correct ; only the BOM divergence needs documenting.
R6	U2.14 (TMR) → GND = safety timer disabled	TMR = NC = 5 h default	Intentional? Otherwise leave NC.
R7	—	PCB_LAYOUT_STATUS.md: “no .kicad_pcb exists” (2026-06-11)	Repo documentation was stale and has been corrected.

4 • Firmware ↔ board pin map

Compared against `firmware-c-next/src/hal_h743/`.

Matches

Function	Firmware	Board
SAI1 FS / SCK / SD	PE4 / PE5 / PE6	pin 3 / 4 / 5
QSPI CLK / BK2_NCS / IO0-3	PB2 / PC11 / PE7-PE10	pin 36 / 79 / 37-40
Amp /SHDN, /MUTE	PB14, PB15	pin 53, 54
MIDI TX	PD5 (USART2)	pin 86
LCD SCK / MOSI / CS	PA5 / PA7 / PA6	pin 29 / 31 / 30
Encoder 1, 2	PA0/PA1 (TIM2), PC6/PC7 (TIM3)	match — A/B swapped on enc 2, so rotation is inverted
MCP23017 GPA0-4 = cells, GPA5 = XSMT, GPA6 = jack, GPA7 = VBUS	IODIRA 0xDF, GPPUA 0x5F	bit-exact match
Backlight = PCA9685 channel 15 → 2N7002	yes	yes

Diverges

Function	Firmware	Board	Note
LCD DC / RES	PC4 / PC5	PD15 / PD14	PC4/PC5 are now encoder 3 on this board
MCP23017 bus	I ² C1 PB6/PB7 (shared with PCA9685)	I²C4 PD12/PD13	PD12/PD13 are TIM4 = encoder 3 in firmware
Encoder 3	PD12/PD13 (TIM4)	PC4/PC5	see I3 — no timer quadrature
Encoder 4	PA8/PA9 (TIM1)	PB0/PB1	see I3 — no timer quadrature
Encoder push 1-3	PE0 / PE1 / PE3	MCP GPB5-7 + PD4	port required

Conclusion: no respin needed for the pin map alone — but a planned firmware port is, with the single exception of I3, which is a hardware limitation.

5 • LED ↔ PCA9685 channel (as built)

Net name	Actual channel		LED
LD0-LD4	0-4	✓	LED6, LED4, LED5, LED3, LED7
LD5	11	✗	LED16
LD6	6	✓	LED9
LD7	10	✗	LED10
LD8	8	✓	LED11
LD9	7	✗	LED13
LD10	5	✗	LED8
LD11	14	✗	LED14
LD12	13	✗	LED15
LD13	12	✗	LED12
LD14	9	✗	LED17 (also reversed — see B5)
LD15	15	✓	backlight FET Q1

6 • Cosmetic and documentation debt

Violations of `AI_READY_SCHEMATIC_STANDARD.md` — not fabrication-critical, but they make the schematic unusable as machine-readable design data.

- **Net names containing spaces:** LCD CS, LCD DC, LCD SCK, LCD RST, SCL MCP, SDA MCP, SCL PCA, SDA PCA, OUT DR+, OUT DR-. Active-low suffixes are missing throughout: SHDN → AMP_SHDN_N, MUTE → AMP_MUTE_N.
- **Net labels contradict the actual pins:** “PB0”/“PB1” on pins 35/34 (swapped), “PC6”/“PC7” on pins 63/64 (swapped), “GPA5” on MCP pin 25 (which is GPA4).
- **Wrong MPN / package fields in the schematic.** The BOM is correct in every one of these cases, so nothing gets mis-ordered — but the schematic lies: R10/R14 read “470 Ω ” with MPN field RC0603FR-07470KL = **470 k Ω** (BOM: C144657 = 470 Ω ✓); R8/R53 read “200 k Ω ” with MPN field 0603WAF3900T5E = **390 Ω** (BOM: C105574 = 200 k ✓); R37/R38/R55/R56 carry footprint field R0402 with an 0603 MPN (BOM: C99782 = 4.7 k 0402 ✓, and the pads really are 0402 ✓); the C39 family uses a library named “10uF 0805” on a C1210 land.
- **Symbol typos:** PAM8403 pin 12 is named “SHND”; the MCP23017 symbol names pin 12 “SCK” (the SPI name of the MCP23S17 — here it is SCL).
- **The BOM lists H2-H5 (mounting holes) as line items with an empty LCSC field.** The JLC BOM upload will error on those. Delete them or mark DNP.
- copper_finish: “None” in the stack-up, and correspondingly “Finish”: “None” in the gerber job file. Set ENIG/HASL so the fabrication data matches the order.
- **Unsaved schematic edits inside the archive:** _autosave-FIELD AMBIENCE.kicad_sch (30 Jul, 14:23) is **newer** than FIELD AMBIENCE.kicad_sch (29 Jul, 22:39) and additionally contains the Mechanical:MountingHole_Pad symbols (H2-H5). The saved schematic is therefore behind the board by exactly those four holes. The .kicad_pcb is identical to its autosave ✓.

7 • What checked out clean

Verified, no objection — recorded so the report is not read as “everything is broken”.

- **BOM ↔ CPL ↔ PCB are 1:1 consistent:** 195 designators in board and BOM, no orphan in either direction, every line quantity equals its designator count, LCSC numbers populated throughout (except H2-H5), CPL correctly holds 191 rows (holes excluded), and top/bottom assignment agrees between CPL and PCB (69 top / 122 bottom).
- **No floating nets:** all 239 nets have at least two pads.
- **No pour shorts:** despite connect_pads (clearance 0) on the GND zone, the measured fill polygon keeps **exactly 0.2 mm** to every foreign pad — 583 pad instances checked, median 0.201 mm.
- **STM32 decoupling is good:** one 100 nF within 1.8–3.3 mm of each of the five VDD pins; VCAP1/VCAP2 with 2.2 μ F at 2.8 / 3.2 mm.
- **BQ24074 correctly configured** (against this project's verified reading of TI SLUS810N): EN1 = LO / EN2 = HI selects resistor-programmed ILIM ✓; CE_N = GND ✓; **ITERM = NC is the datasheet default (10 %) and therefore correct** ✓; ISET 1.13 k $\approx$ 0.79 A; ILIM 1.18 k $\approx$ 1.36 A; EP on GND ✓; /CHG and /PGOOD LEDs correctly polarised ✓.
- **TPS61089:** FB divider 332 k / 107 k → 4.94 V ✓; **R_FSW returning to the SW node is correct** per TI SLVSD38C table 6-1 — it looks wrong but is not ✓; BOOT capacitor present ✓.
- **TPS22918:** QOD tied to VOUT, CT left open — exactly the ADR-0016 intent ✓.
- **TPA6132A2:** charge-pump network complete (CPP/CPN/HPVDD/HPVSS), single-ended inputs with IN+ to GND, EP on GND ✓.
- **PCM5102A / APS6404L / MCP23017 / PCA9685:** pin assignments checked against the symbols, no transposition.
- **USB ESD (USBLC6-2SC6) present, CC1/CC2 with 5.1 k ✓.**
- **Board outline closed,** 198.84 × 106.92 mm, 4 layers, 1.6 mm, vias 0.6/0.3 mm, inner-layer clearance 0.127 mm — all within standard JLC capability.

8 • Recommended order of work

	Items	Action
1	B1 + B2 + B3 together	Redraw and re-place the power stage: inductor, switch path, pull-down, input capacitor at the IC.
2	B4, B5, B7	Pure netlist corrections.
3	B6	Firmware one-liner (MODE2 = 0x00) — can be done now.
4	R1 + R2 + R4	Bring the audio front end up to the r19.37 state and verify the LDO pinout.
5	I1, I2, I14, I15	Four small netlist additions.
6	I3 + I4	Straighten out the encoder pins, then plan the firmware port.
7	I6	Layer stack to SPEC §9.
8	I7 + I8	Restore DRC severities, fix NPTH, re-export gerbers — only then is a DRC run meaningful.

Three items could not be verified from the available sources and are marked **UNVERIFIED — NEEDS HUMAN CHECK** per the project's schematic standard: the PJ-320D pad-to-contact mapping, the AP7361C SOT-223 pin order, and the exact display module behind H1.